

Module 13: How Populations Evolve — Questions

Quick Check

- 1. Microevolution is officially defined in biology as:** A) A change in allele frequencies in a population over generations. B) The sudden appearance of a brand new species. C) The process where an individual animal adapts to a new habitat. D) The extinction of a population due to disease.
 - 2. A large population of lizards is dramatically reduced by a massive flash flood. The few surviving lizards happen to be overwhelmingly green, whereas the original population was a mix of green and brown. The future population will rebuild, but it will be mostly green. This is an example of:** A) The Founder Effect B) Gene Flow C) The Bottleneck Effect D) Disruptive Selection
 - 3. Which of the following is the ONLY mechanism of microevolution that consistently results in populations becoming better adapted to their environments?** A) Genetic drift B) Mutation C) Natural selection D) Gene flow
 - 4. A population of plants produces seeds that are eaten by birds. Extremely heavy seeds are too hard for birds to carry, and extremely light seeds blow away in the wind and die. Over time, the population almost exclusively produces medium-weight seeds. What type of selection is this?** A) Directional Selection B) Stabilizing Selection C) Disruptive Selection D) Sexual Selection
 - 5. True or False:** A population in Hardy-Weinberg equilibrium is actively evolving. - ☐ True
- ☐ False
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Reflection

- 6. Explain the core difference between Gene Flow and Genetic Drift.**

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7. African cheetahs are incredibly similar genetically, to the point where skin grafts between unrelated individuals are rarely rejected. Scientists believe thousands of years ago, widespread climate changes nearly wiped them out, leaving only a tiny handful of survivors to repopulate the species. Which specific evolutionary mechanism does this describe, and why is this lack of genetic diversity a severe danger for cheetahs today?

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8. Draw or describe a scenario that represents Disruptive Selection in an animal population. Why are the extreme traits favored over the average trait?

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